

HAKADAL, Norway

WMO: 014880

Lat: 60.1172N

Lon: 10.8289E

Elev: 170

StdP: 99.30

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-16.7	-21.4	0.6	-18.7	-18.7	0.7	-16.1	8.6	2.5	7.1	1.8	0.9	290	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	27.0	17.4	25.1	16.8	23.1	15.8	19.1	23.6	18.1	23.0	17.0	21.5	2.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.0	20.6	16.2	11.8	19.4	15.2	11.0	18.5	55.0	23.5	51.7	22.9	48.3	21.6	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.6	4.5	-21.7	29.1	3.1	2.5	-23.9	30.9	-25.7	32.3	-27.4	33.7	-29.6	35.5		
(5)			-21.9	20.0	3.1	1.9	-24.1	21.3	-25.9	22.4	-27.6	23.4	-29.9	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-5.3	-4.1	0.1	4.8	10.3	14.3	17.1	15.7	11.2	5.5	0.4	-4.2
	DBStd	8.76	5.34	5.25	3.92	3.14	3.89	2.74	2.81	3.13	2.93	3.48	4.06	5.73
	HDD10.0	2272	474	394	306	159	44	2	0	0	21	145	287	441
	HDD18.3	4707	733	627	564	406	249	126	59	97	214	398	537	699
	CDD10.0	645	0	0	0	3	54	130	219	176	57	6	0	0
	CDD18.3	39	0	0	0	0	1	4	21	14	0	0	0	0
	CDH23.3	377	0	0	0	0	22	66	210	78	1	0	0	0
	CDH26.7	60	0	0	0	0	2	10	41	7	0	0	0	0
(14) Wind	WSAvg	1.6	1.4	1.5	1.8	1.9	1.8	1.8	1.7	1.5	1.6	1.5	1.3	1.3
(15) Precipitation	PrecAvg	875	73	48	41	55	62	74	90	91	80	98	102	60
	PrecMax	1272	162	95	126	109	105	166	177	188	155	249	328	100
	PrecMin	694	14	18	0	0	5	7	13	24	23	18	24	10
	PrecStd	131	38	22	32	27	33	42	38	41	29	50	68	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	8.2	13.7	19.4	26.0	28.0	30.0	27.7	21.4	15.3	11.3	8.0
		MCWB	4.6	4.3	7.7	10.9	14.7	16.8	18.2	18.8	15.2	11.6	9.5	6.2
	2%	DB	4.0	5.9	10.3	15.8	23.0	25.2	27.8	25.6	19.4	13.8	8.9	6.2
		MCWB	3.0	3.2	5.2	9.1	13.9	15.4	17.6	18.0	13.9	11.3	7.8	5.2
	5%	DB	2.8	4.2	7.7	13.1	20.7	22.7	25.8	23.7	17.9	12.8	7.5	4.7
		MCWB	1.9	2.7	3.4	7.5	12.9	14.3	17.2	17.1	12.9	10.7	6.6	3.7
	10%	DB	1.3	2.9	5.8	11.0	17.9	20.5	23.8	21.7	16.4	11.5	6.2	2.9
		MCWB	0.6	1.7	2.7	6.3	11.6	13.1	16.5	15.9	12.9	9.5	5.1	2.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	5.7	7.9	12.0	16.6	17.9	20.2	20.5	16.9	13.6	10.0	6.8
		MCDB	5.6	7.4	13.0	17.3	21.8	23.8	24.8	24.0	19.8	14.6	10.8	7.4
	2%	WB	3.2	4.0	5.7	9.9	15.0	16.6	19.1	19.4	15.4	12.2	7.9	5.3
		MCDB	3.7	4.9	8.8	14.3	20.5	22.3	24.3	23.2	17.3	13.1	8.5	6.1
	5%	WB	1.9	2.9	4.2	8.5	13.6	15.6	18.2	18.3	14.4	11.2	6.8	3.7
		MCDB	2.3	3.9	6.6	11.9	18.7	20.9	23.7	22.3	16.8	12.4	7.5	4.4
	10%	WB	0.7	1.7	3.1	7.1	12.5	14.5	17.3	17.0	13.4	9.7	5.3	2.0
		MCDB	1.2	2.7	5.3	10.4	17.2	18.9	22.4	20.6	15.7	11.0	6.0	2.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	7.0	8.9	10.1	10.7	10.1	10.6	9.9	8.9	7.1	4.7	5.5
		MCDBR	5.7	7.5	11.7	14.2	15.5	14.4	14.4	11.5	11.1	8.0	6.9	6.8
		MCWBR	4.8	5.7	7.2	8.0	7.5	6.5	6.1	5.2	6.5	5.8	6.0	6.3
	5% WB	MCDWR	5.0	6.6	10.3	12.3	12.4	11.6	11.9	9.9	8.5	6.9	6.4	6.5
		MCWBR	4.6	5.4	6.8	7.3	6.5	6.0	5.6	5.1	5.6	5.7	5.8	6.0
(40) Clear-Sky Solar Irradiance	taub		0.255	0.297	0.325	0.369	0.366	0.355	0.376	0.369	0.335	0.321	0.275	0.227
	taud		2.194	2.272	2.313	2.305	2.397	2.448	2.409	2.425	2.506	2.453	2.279	2.105
	Ebn at Noon		527	704	803	827	861	876	846	823	796	680	500	402
	Edh at Noon		40	67	88	107	105	101	103	94	74	57	37	28
(44) All-Sky Solar Radiation	RadAvg		0.28	0.90	2.37	3.76	4.85	5.55	4.96	3.90	2.58	1.14	0.38	0.18
	RadStd		0.05	0.16	0.25	0.50	0.50	0.43	0.65	0.32	0.30	0.22	0.10	0.02

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (56 neighbors)		N/A	N/A	+1.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page